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Project Title:	Histone Acetyltransferases and the Cellular Response to DNA Damage		
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Task Description:	This study will focus on the Space Radiation Program Element (F.1) dose and dose-rate effects from Solar Particle Events (SPE) and the Bioastronautics Roadmap radiation health risk Carcinogenesis, Research and Technology Questions 28i and 28j. Exposure to SPEs and Galactic Cosmic Ray (GCR) particles cause DNA lesions - oxidation, abasic sites, single- and double-strand breaks, or combination of lesions called cluster damage. Their proper and timely repair prevents the accumulation of mutations that could promote cellular oncogenic transformation. Human DNA is organized into a tight, high-order structure, chromatin, with basic building blocks, nucleosomes, in which the DNA is wrapped around a core particle of proteins called histones. During repair, this structure must be opened up to allow the access of repair enzymes to DNA. Repairing different damages entails unpacking chromatin to different degrees, with complex cluster damages requiring major rearrangements. I will focus on the involvement of enzyme called histone acetyltransferase 1 (HAT1) in DNA repair. Most HAT enzymes are part of network that facilitates the replication or the transcription by acetylation of histone molecules that regulate chromatin assembly and accessibility. At least one of them, TIP60, also plays a major role in DNA repair. I will explore the ability of HAT1 to adequately open the chromatin after specific DNA damage so to promote efficient repair, particularly evaluating the extent to which its activation depends on the type, the dose, and the dose-rate of the damaging agent. The possibility that HAT1 acts as structural protein also will be assessed. A model will be created relating HAT 1's activity in DNA repair, and changes in cell growth toward malignancy. My data ultimately will help in accomplishing the final goal - reducing the risk of carcinogenesis to astronauts during space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research provides insights into the molecular changes in response to DNA damages inflicted by protons, HZE and secondary particles in human cells. We introduced a new type of normal human cell, viz., keratinocytes, to study DNA damage. We believe they are important for future studies because during space flights, skin keratinocytes would be exposed to higher doses of radiation than other cell types, and they have the potential for malignant transformations. Using the expression and distribution of Hat1 as a tool, our findings will allow us to assess the effect of the radiation's quality and the dose-rate dependence of SPEs and GCRs on humans during space exploration, thereby to predict the carcinogenic risk from these types of radiation and mitigate the consequences. Our results also are applicable in the evaluation of the DNA damage to the skin of cancer patient from treatment with X-rays and HZE particles because the ionizing radiation first will penetrate the skin of the body and then reach the potential target.
Task Progress:	Human HAT1 isoform a has an exclusively nuclear localization The "NCBI Reference Sequences" database shows that two transcriptional variants originate from the human HAT1 gene. Isoform a codes for a protein that has 419 amino acids (aa), while isoform b codes for one with 334 aa. The latter uses an internal methionine for its start codon, and lacks the first 85 aa compared to variant a. Our first goal was to determine the expression pattern of Hat1 and to establish which of the two transcriptional variants is expressed in human keratinocytes. We chose to use normal human keratinocytes (NHK) because they have a well-defined cytoplasm and nucleus. The cells were grown in serum free media and protein extract were prepared. Western blotting analysis using anti–Hat1 antibody, clone EE21, that recognizes the Hat1 C-terminal domain, thus simultaneously detecting both Hat1 a and b. The membranes were then striped and re-probed with anti–beta-actin and anti–Lamin A antibodies. We confirmed that Hat1 is located in the cytoplasm and the nucleus of normal human cells. However for first time we show that it is expressed in two isoforms, a and b. While isoform a is confined to the nucleus of normal human cells, isoform b is distributed throughout the cytoplasm and the nucleus. Isoform a, and select parts of isoform b are insoluble and associate with the cells' nuclear matrix. Immunofluorescent analysis of Hat1 distribution in NHKs after treating with DNA damaging agents We explored the participation of human Hat1 in the cellular responses of NHKs to treatment with hydrogen peroxide (H2O2), gamma rays, or a beam of 1000 MeV/n iron ions. The treated cells were fixed and immunostained to reveal the distribution of Hat1 a and b. Hat1 occurs in the nuclei and cytoplasm of untreated NHK. Mitotic cells stain brightly for Hat1, while certain cells not undergoing mitotic division also show brighter nuclear staining, presumably because they are in the S-phase of the cell cycle. Twenty minutes after exposing the keratinocytes to 0.5 mM H2O2, we observed a notable change in the staining pattern for Hat1. Cells that were in late telophase of mitotic division still stain brightly for Hat1. In certain cells Hat1 is generally absent from the nucleus. Besides these exceptions, almost the entire population of cells shows brighter nuclear staining for Hat1. Thus, although the cytoplasm is still noticeably stained for Hat1, the nuclei have experienced a 5- to 10-fold increase in the Hat1 signal, as shown on the surface plot for Hat1. Further, it looks as though the staining for Hat1 now extends throughout the entire nucleus of the H2O2 treated cells; however further studies are needed to prove it unequivocally. Exposure of NHKs to 50 cGy of gamma radiation did not significantly change the distribution of cellular Hat1. In contrast, 15 min after the exposing NHKs to 20 or 50 cGy of beam of 1000 MeV/n of iron particles, the distribution of Hat1 was similar to that apparent in cells exposed to H2O2. More than 90% of these cells exhibit bright nuclear staining for Hat1. However, the nuclei of cells expressing senescent-like phonotype do not stain for Hat1. Cells undergoing mitotic division reveal a disordered pattern of chromosomal DNA but still stain brightly for Hat1. We observed the beginning of enhanced nuclear staining in some cells just 5 minutes post-irradiation. By 15 to 20 minutes, enhancement was evident in more than 80% of the cells, and persisted for more than 4 hours; thereafter, it disappeared within 18 – 24 hours. This correlates well with the rate of the DNA repair evaluated by the formation and dissolution of the p53 binding protein 1 (53BP1) foci at the site of the damage. We performed double immunostaining for Hat1 and for 53BP1 proteins in unirradiated and cells irradiated with 20 cGy of 1 GeV/n iron ions. The staining showed that 20 min after irradiation 20 % of the cells express 53BP1 foci, and 24 hours post-irradiation, the number of cells with 53BP1 foci gradually dropped to background levels. The enhanced nuclear staining for Hat1 is most likely due to an intra-nuclear redistribution. This radiation and H2O2 induced redistribution could entail Hat1's physical attachment to the chromatin, or its attachment to the nuclear lamina, thus obscuring the actual distribution of Hat1 inside the nucleus. Normal human keratinocytes (NHKs) are appropriate system for studying DNA damage Several cell types and cell lines have been used to study radiation-induced DNA damage in human cells. We chose NHKs that do not show radiation hypersensitivity. The astronaut's skin is the organ that will receive the highest doses of radiation during space travel; the internal organs receive 2- to 3-times less due to self-shielding by the body, especially after exposure to SPE protons. The epidermis consists of several layers of keratinocytes with different proliferative abilities. The basal layer contains

keratinocyte stem cells with limited proliferative ability, a characteristic that makes NHKs useful for in vitro studies. As a surrogate marker for DSBs we used 53BP1 foci that form at the sites of the breaks in the irradiated cells that also allowed us to enumerate the extent of the damage. At the sites of the DSBs, 53BP1 co-localizes with the gamma-H2AX foci, and can serve to visualize DSB, even at very low levels.

For their normal physiological growth, NHKs require retinol that they convert to a variety of metabolites, including retinoic acid (RA). RA is an important co-factor in the transcription regulation of several nuclear genes, and regulates the cellular differentiation in several cell types. It is involved, through various pathways, in activating 53BP1, an ATM substrate that, upon phosphorylation by this and other kinases, forms foci at the site of DNA DSBs.

The formula of the keratinocyte serum-free media does not contain retinol. We designed our first set of experiments to establish whether the presence of the PI3K inhibitor wortmannin, which inhibits ATM and other kinases, and the presence of retinol in the growth media affect the formation of the 53BP1 foci in response to ionizing radiation. NHKs were grown on cover slides then irradiated with 20 cm X 20 cm square beam of 1 GeV/n iron particles. After irradiation, the cells were left to recover then fixed and stained to visualize the 53BP1 distribution.

Supplementing the growth media with retinol builds up the cells' internal pool of retinoids. This addition dramatically affected the NHKs exposed to 2.5 mGy of 20 cm X 20 cm square beam of 1 GeV/n iron ions. There was a two-fold increase in the number of cells with 53BP1 foci compared to cells grown without retinol; this increase was eliminated completely by pre-incubating the cells with wortmannin. NHKs lacking an internal build up of retinoids failed to show a rise in 53BP1 foci formation after exposure to the 1 GeV/n iron beam in the DMSO and in the wortmannin treated samples. Accordingly, we concluded that the formation of 53BP1 foci depends on PI3K signaling and requires an internal pool of retinoids.

Exposure of NHK to secondary particles produced by the interaction of 1 GeV/n iron beam with an aluminum block

Earlier simulations support our findings that interaction of iron particles with an aluminum block generates secondary particles with lower atomic number and energy. These particles disperse over a broad angular range, delivering radiation doses far from the initial point of collision. We evaluated whether these secondary particles would create DNA damage similar to that observed after exposure to a beam of 1 GeV/n iron particles. The cells were exposed to "low" and "high" doses of secondary particles differing by a factor of ten. Exposing the NHKs to secondary particle radiation from the interaction of iron ions with the aluminum block induced DNA damage, revealed as 53BP1 foci. The number of induced 53BP1 foci depends on the dose and the location of the samples. Like irradiation with HZE particles, exposing NHKs to secondary particles induced foci formation that depends on the presence of an internal pool of retinoids, and is hampered by the PI3K inhibitor, wortmannin.

HZE particles cause clustered DNA damage that is persistent, and very difficult to repair. After exposure, the cells were supplemented with media with retinol and allowed to grow for 20 minutes, 1, 2, 4, or 24 hours, at which points they were processed for immunostaining to detect the 53BP1 protein. NHKs exposed to the low- and the high-dose of secondary particles developed significantly higher numbers of foci than the controls. In both cases, this increase was apparent at 20 minutes after exposure. The number fell with time; nevertheless, at 24 hours, significant fractions of the keratinocytes still contained foci. Apparently, the damages were not sufficiently severe to trigger apoptosis of the damaged cells. We concluded that the secondary particles inflicted severe, persistent DNA damages to human cells and could not be repaired readily.

Bibliography Type:	Description: (Last Updated: 03/11/2021)
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